

B. Tech Examination 2024-25

(Odd Semester)

COMPUTER CONCEPTS & PROGRAMMING IN C

Time : Three Hours]

[Maximum Marks : 60

Note :— Attempt all questions.

SECTION-A

1. Attempt all parts of the following : $8 \times 1 = 8$

- (a) What do you mean by flowchart?
- (b) Give the rules for variable declaration.
- (c) What is the use of sizeof () operator?
- (d) Differentiate between break and continue.
- (e) What is recursion?

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- (f) Write some of the differences between structure and union?
- (g) Why do we use header files?
- (h) What is the full form of EOF?

SECTION – B

2. Attempt any two parts of the following: $2 \times 6 = 12$
- (a) What are the different data types available in "C"? How many bytes are occupied by them?
 - (b) Explain the difference between bitwise AND and logical AND operators with the help of examples.
 - (c) Write a C program to print the sum of diagonal elements of 2-D matrix.
 - (d) Explain different categories of pre-processor directives used in C.

SECTION – C

Note:- Attempt all questions. Attempt any two parts from each question. $5 \times 8 = 40$

- 3. (a) What is an operating system? Mention the types of operating systems and explain any two of them in detail.
- (b) Explain the basic structure of a "C" program.
- (c) Explain the significance of an assembler and compiler.
- 4. (a) What are different bit operators used in C? Give an example of each.
- (b) Write a program to print the count of even numbers between 1 to 200. Also print their sum.
- (c) What are iterative control statements? Explain the difference between while loop and do while loop.
- 5. (a) How will you pass parameters to a function? Briefly describe two mechanisms of parameter passing in "C" language.
- (b) What is an array? How a single dimension and two dimension arrays are declared and initialized?

- (c) Write a C program to read and display student details using structure.
6. (a) How a pointer variable is declared? What is the purpose of data type included in the declaration?
- (b) Explain the following preprocessor directives?
- (i) #include
 - (ii) #define
- (c) Write a short notes on:
- (i) fgets()
 - (ii) fputs()